What Horizon-Aware Losses Do in Multi-Step Forecasting: The L1 Norm, Not Decorrelation

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Abstract

Several recent losses for direct multi-step forecasting, including FreDF, Time-o1 and QDF, are motivated by the claim that autocorrelation among the future labels biases the mean squared error (MSE). We show that this claim does not hold for quadratic or orthogonally transformed losses, which share the population minimizer of the MSE and, for least squares on a shared design, its solution; re-weighting the output directions only redistributes regularization, with signal-to-noise ratios rather than the inverse noise covariance as optimal weights. A controlled decomposition of the official FreDF and Time-o1 code shows that their gains, in MSE and in MAE, come from the L1 norm, since L2 versions in the same bases match or trail the MSE. The size of these gains depends mainly on whether the model normalizes each input window: without normalization, the L1 norm keeps the largest training errors from dominating the fit, and in a linear model, trimming the largest-error windows from the MSE recovers 42% to all of the gain, whereas clipping the global gradient norm recovers little. With normalization, a smaller gain remains on iTransformer, mostly in the forecast level; its cause is open, as is the mechanism of QDF on deep models, since we test output-direction weighting only with linear heads. Under rolling origins, whether the frequency basis helps depends on the backbone, and choosing between the MSE and a robust loss on a validation period is unreliable because the gains vary with the season and with level drift.

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1. Introduction

Direct multi-step forecasters predict the next H values of a series from a lookback window in a single pass, and they are usually trained with the mean squared error (MSE) over the horizon. Several recent works argue that this objective is biased because it ignores the autocorrelation among the H future values. FreDF aligns forecasts and labels in the frequency domain (Wang et al., 2025b), Time-o1 aligns them in a principal-component basis of the labels (Wang et al., 2025a), and QDF learns a quadratic weight matrix that accounts for the correlation (Wang et al., 2026a). DistDF and a survey by the same authors extend this line of work (Wang et al., 2026b;c). These losses report consistent improvements over the MSE on standard benchmarks.

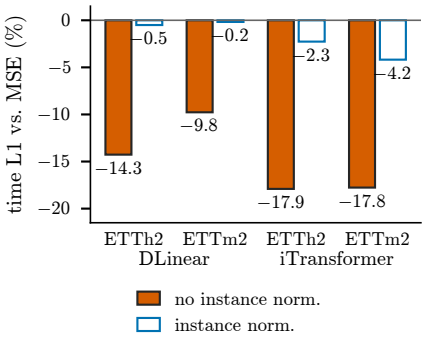
This motivation conflicts with a basic property of the squared error, whose minimizer is the conditional mean regardless of how the coordinates of the target are correlated. Nevertheless, when we re-run the official code, FreDF improves on the same script trained with the MSE by 4.1–4.5% on the standard split of ETTh1, ETTh2 and ETTm1 (Section 5.1) and by 2.0–5.4% on average over five rolling origins of the four ETT datasets (Section 5.2), so the reported gains are reproducible. In this paper, we study what these losses change in practice and when they help. That quadratic losses share the minimizer of the MSE is classical, and An et al. (2026) have already questioned the decorrelation argument of FreDF. Our main contribution is therefore empirical: we decompose the official losses into a transform and a norm, relate their gains to instance normalization, and test competing explanations by interventions on the training loss.

Quadratic losses, and L2 losses after an orthogonal transform, have the same population minimizer as the MSE, also for misspecified model classes that use the same function class for every horizon step, and with a shared design matrix they give the same least-squares solution (Section 3). In the official implementations, the gains of FreDF and Time-o1 come from their L1 norm, and they are largest when the model does not normalize its input windows. There, they come from bounding the influence of the largest training errors, which otherwise dominate a fit that transfers badly to a drifted test period; with normalization, a

(a) Training loss = transform of the forecast error, then a norm

	Time domain ($T = I$)	Fourier (real FFT)	Label PCA
L2 norm	MSE (reference)	FFT-L2: 0.0% rFFT-L2: +0.4 to +3.9%	PCA-L2: +0.1 to +1.7%
	same population minimizer as the MSE (Prop. 1, Thm. 1)		
L1 norm	time L1: −2.0 to −5.4% Huber: −1.5 to −5.3% window $L_{2,1}$: −1.2 to −4.7%	freq. L1: −4.0 to −4.6% FreDF: −2.7 to −4.5%	PCA-L1: −0.6 to −4.1% Time-o1: −0.6 to −4.2%
	same target as MSE for symmetric noise (Prop. 3); gain depends on normalization, see (b)		

(b) L1 gain with and without instance normalization



Frequency vs. time L1 under rolling origins (Table 2): −0.1 to −1.5% on iTransformer, +0.8 to +4.4% on DLinear

Figure 1. The losses studied in this paper, written as a transform of the forecast error followed by a norm. (a) Columns give the transform and rows the norm (grey: L2, blue: L1). Each cell lists the losses of that type with the range, over ETTh1, ETTh2, ETTm1 and Weather, of their paired change in test MSE relative to the same script trained with the MSE (standard split; iTransformer for the FreDF script, Fredformer and FreTS for the Time-o1 script; Figure 2). L2 losses in an orthonormal basis share the minimizer of the MSE and do not improve on it, while L1 losses carry the gains. FreDF and Time-o1 (framed) are L1 losses in a rotated basis. The bottom line gives frequency versus time L1 under rolling origins (Table 2). (b) Change in test MSE of time L1 over the MSE with and without instance normalization on ETTh2 and ETTm2 (Figure 3); in their official settings DLinear runs without and iTransformer with normalization.

smaller gain remains, mostly in the forecast level, whose cause we could not establish (Sections 5.1–5.3). Figure 1 summarizes the decomposition and the role of normalization. Whether the frequency basis helps relative to the time domain depends on the backbone, and choosing the loss on a validation period fails when that period differs from the deployment period in season or level (Section 5.5).

Our contributions are as follows.

- We show that weighted and orthogonally transformed L2 losses are equivalent to the MSE in the population and, with a shared design, in finite samples, also for misspecified classes that share one function class across horizon steps. Weighting acts as a separate ridge penalty for each output direction with the signal-to-noise ratios as optimal weights, which predicts when GLS-type weights hurt and why weighting changes almost nothing on our data (Sections 3 and 5.4). Under symmetric noise, transformed L1 losses also target the conditional mean.
- Using the official code, we show that L2 variants of FreDF and Time-o1 match or trail the MSE, while L1 variants reproduce their gains (Section 5.1).
- We show in a linear model, DLinear and iTransformer that the size of the L1 gains depends mainly on instance normalization. Interventions on the training loss of the linear model show that, without normalization, the gains come from keeping the largest training errors from dominating the fit. With normalization, most of the remaining gain on iTransformer is a smaller error in the forecast level (Section 5.3).
- We evaluate on rolling origins with Diebold–Mariano tests whose size we check by simulation. The frequency basis helps iTransformer but hurts DLinear, and choosing between the MSE and a robust loss on validation data is unreliable (Sections 5.2 and 5.5).

2. Setting and Loss Families

We consider direct multi-step forecasting, in which one model outputs all H future values at once (Chevillon, 2007; Ben Taieb et al., 2012; Ben Taieb & Atiya, 2016). We observe windows $(x_i, y_i)_{i=1}^n$, where x_i is the lookback and $y_i \in \mathbb{R}^H$ contains the next H values of the same series; consecutive windows overlap. Let $m(x) = \mathbb{E}[y \mid x]$, $\varepsilon = y - m(x)$ and $\Sigma_\varepsilon = \mathbb{E}\,\varepsilon\varepsilon^\top$. Forecasts are evaluated by $\mathrm{MSE}(f) = \frac{1}{H}\mathbb{E}\|y - f(x)\|^2$. We consider two families of training losses and their mixtures. The first is the quadratic loss $\ell_\Omega(\hat{y}, y) = (y - \hat{y})^\top \Omega (y - \hat{y})$ with $\Omega \succeq 0$. The second is the transformed group-L1 loss $\ell_{T,\mathcal{G}}(\hat{y}, y) = \sum_{g \in \mathcal{G}} \|(T(y - \hat{y}))_g\|_2$, where T is a linear map and $\mathcal{G}$ is a partition of its rows. Table 1 shows how the losses we study fit into these two families.

An orthonormal transform followed by an L2 norm leaves the loss unchanged, as for the full-spectrum FFT and a full-rank PCA, so the transforms used by FreDF and Time-o1 can matter only through re-weighting or through the L1 norm. In their official settings, both methods are L1 losses in a rotated basis, up to a small MSE term and a diagonal standardization.

Table 1. Losses in the framework ($e = \hat{y} - y$; $K = \lfloor H/2 \rfloor + 1$; $a_t = (-1)^t$; $\mathbf{1}$ the all-ones vector). The rFFT form holds for even H (Appendix A.1).

Loss	Form
MSE (direct forecasting)	$\Omega = I/H$
Full-spectrum FFT L2	$\Omega = I$ (Parseval)
rFFT half-spectrum L2	$\Omega = \frac{H}{2K}\left(I + \frac{1}{H}(\mathbf{1}\mathbf{1}^\top + aa^\top)\right)$: DC and Nyquist weighted twice
FreDF	$\lambda\,\mathrm{MSE} + (1-\lambda)\frac{1}{K}\sum_k \lvert\mathrm{rFFT}(e)_k\rvert$: quadratic + group L1, groups = (Re, Im) pairs
Time-o1	per-step standardization, then L1 of label-PCA coefficients (optionally truncated)
QDF	learned Ω, motivated by $\Omega \approx \Sigma_\varepsilon^{-1}$
Time L1 / Huber	$T = I$, singleton groups / Huber
Window $L_{2,1}$	$T = I$, one group per window

Related work. Beyond the losses above, training objectives for forecasting include shape- and structure-aware losses (Le Guen & Thome, 2019; Kudrat et al., 2025; Qiu et al., 2025), constrained losses per horizon step (Hounie et al., 2024), and learnable or distributional losses (Wang et al., 2026d;b). Shi et al. (2025) and Cai et al. (2025) argue that unitary transforms reduce an optimization bias of point-wise losses. The work closest to ours is An et al. (2026), who point out that FreDF's decorrelation argument is statistically inaccurate, because the DFT is linear and preserves the correlation structure and, by Parseval, the time- and frequency-domain MSE differ only by a constant. They attribute the effectiveness of the frequency-domain MAE to the shape of its gradient. FreDF itself motivates its L1 norm by optimization as well: the frequency components differ greatly in magnitude, which makes the squared loss unstable (Wang et al., 2025b). We extend the observation of An et al. (2026) to weighted losses, misspecified models and finite samples. Empirically, our contribution is not that an L1 loss can beat the MSE: we decompose the official losses into transform and norm, relate their gains to instance normalization, test the optimization and robust-estimation accounts by interventions on the training loss, and evaluate validation-based loss selection across origins.

Several of our results restate classical facts in the forecasting setting. With identical regressors, least squares does not depend on the noise covariance (Zellner, 1962; Kruskal, 1968), and correlation between the responses can help only through constraints or shrinkage, as in reduced-rank regression (Anderson, 1951; Izenman, 1975; Reinsel & Velu, 1998) and multi-response shrinkage (Breiman & Friedman, 1997). Our reading of weighting as a ridge penalty per output direction uses spectral analyses of ridge regression and early stopping (Ali et al., 2019; Dobriban & Wager, 2018; Hastie et al., 2022; Wu & Xu, 2020). We take an exact linear least-squares fit as a reference because linear forecasters are strong baselines with closed-form solutions (Zeng et al., 2023; Li et al., 2023; Xu et al., 2024; Toner & Darlow, 2024), and the mechanism experiments vary instance normalization, the usual remedy for level shifts (Kim et al., 2022; Berthelier et al., 2026). The rolling-origin evaluation follows work on out-of-sample testing and benchmarking (Tashman, 2000; Bergmeir & Benítez, 2012; Hewamalage et al., 2022; Qiu et al., 2024), and the tests follow work on comparing predictive accuracy when errors are dependent (Diebold & Mariano, 1995; Lazarus et al., 2018; Coroneo & Iacone, 2020).

3. What Weighted and Transformed Losses Can Change

The proofs are in Appendix A.

3.1. Population Level

Proposition 1 (Quadratic losses)**.** *Let $\Omega \succ 0$ and $R_\Omega(f) = \mathbb{E}\|y - f(x)\|_\Omega^2$. Then $R_\Omega(f) = \mathbb{E}\|\varepsilon\|_\Omega^2 + \mathbb{E}\|m(x) - f(x)\|_\Omega^2$. Hence (a) if $m \in \mathcal{F}$, m minimizes R_Ω over $\mathcal{F}$ for every $\Omega \succ 0$; (b) for any class $\mathcal{F}$, a minimizer f_Ω of R_Ω over $\mathcal{F}$ has $\mathrm{MSE}(f_\Omega) \geq \min_{f\in\mathcal{F}} \mathrm{MSE}(f)$.*

Proposition 1 implies that weighting the horizon cannot correct a bias of the MSE; at best it trades the MSE for another criterion. If Ω is singular, as after truncating a basis, the components in $\ker\Omega$ are left undetermined. A zero-initialized linear head then predicts 0 in these directions, which gives an irreducible bias $\mathbb{E}\|P_{\ker\Omega}(m - f)\|^2$.

Proposition 2 (Shared function class across horizon steps)**.** *Let $\mathcal{F} = \mathcal{G}^H = \{(f_1, \ldots, f_H) : f_h \in \mathcal{G}\}$ with $\mathcal{G}$ a closed linear subspace of $L^2(P_x)$, e.g. an unconstrained linear head on fixed features, which covers DLinear, RLinear and FITS (Toner & Darlow, 2024). Then $\arg\min_\mathcal{F} R_\Omega = \{\Pi_\mathcal{G} m\}$, the coordinate-wise projection, for every $\Omega \succ 0$, also when $m \notin \mathcal{F}$.*

When every horizon step is fitted with the same function class, weighting therefore does not change the solution, even if the model is misspecified.

Proposition 3 (Transformed L1)**.** *Let $\psi(u) = \sum_g \|(Tu)_g\|_2$. If $\varepsilon \mid x$ is centrally symmetric, then $m(x)$ minimizes $\mu \mapsto \mathbb{E}[\psi(y - \mu) \mid x]$. Hence over all measurable f, transformed L1 losses and the MSE have a common population minimizer.*

The assumption is strong: the noise must be centrally symmetric as a vector given x, and the proposition does not establish uniqueness. Under skewed noise, the minimizer is a (spatial) median of the transformed coordinates and generally differs from m, so the L1 loss is biased for the MSE target. In a simulation with $n = 10^5$ and an exact linear

fit, Gaussian and t_3 noise leave the test MSE of time and frequency L1 at that of least squares, whereas centred exponential noise raises it by 3.6% for time L1 and 1.4% for frequency L1 (Appendix B).

3.2. Finite Samples: Weighting Acts as Regularization per Output Direction

Let the design matrix $\Phi \in \mathbb{R}^{n\times d}$ be shared by the H outputs, $Y \in \mathbb{R}^{n\times H}$, $S = \Phi^\top\Phi/n = U\mathrm{diag}(s)U^\top$ and $\Omega = V\mathrm{diag}(\omega)V^\top$. Write $\tilde{B} = U^\top BV$ and $\tilde{B}^{\mathrm{OLS}} = \mathrm{diag}(s)^{-1}U^\top(\Phi^\top Y/n)V$ (on $s_j > 0$).

Theorem 1. *(a) For every $\Omega \succ 0$, the minimum-norm minimizer of $\mathrm{tr}\left((Y-\Phi B)\Omega(Y-\Phi B)^\top\right)$ is Φ^+Y, the OLS solution. (b) The minimizer of $\frac{1}{2n}\mathrm{tr}\left((Y-\Phi B)\Omega(Y-\Phi B)^\top\right) + \frac{\lambda}{2}\|B\|_F^2$ satisfies $\tilde{B}_{jk} = \frac{s_j\omega_k}{s_j\omega_k+\lambda}\tilde{B}^{\mathrm{OLS}}_{jk}$. (c) Gradient descent on the weighted loss from $B=0$ with step η gives $\tilde{B}_{jk}(t) = \left[1-(1-\eta s_j\omega_k)^t\right]\tilde{B}^{\mathrm{OLS}}_{jk}$.*

Part (a) is the classical result for seemingly unrelated regressions with identical regressors (Zellner, 1962; Kruskal, 1968). Parts (b) and (c) show that each output direction v_k is fitted with its own ridge parameter λ/ω_k or, under gradient descent, its own training time $\omega_k t$. Rotating the outputs without re-weighting therefore leaves the fit unchanged, and truncation corresponds to $\lambda_k = \infty$.

Risk. Suppose $y = B_*^\top\phi(x) + \varepsilon$ with independent rows, noise independent of the design and $\mathrm{Cov}(\varepsilon) = \Sigma_\varepsilon$; let $\beta = U^\top B_* V$ and $\tilde{\sigma}_k^2 = v_k^\top\Sigma_\varepsilon v_k$. The fixed-design excess risk of (b) is $R(\Omega,\lambda) = \sum_k R_k(\lambda/\omega_k)$ with

$$R_k(\lambda_k) = \sum_j \left[s_j\left(\tfrac{\lambda_k}{s_j+\lambda_k}\right)^2\beta_{jk}^2 + \tfrac{\tilde{\sigma}_k^2}{n}\left(\tfrac{s_j}{s_j+\lambda_k}\right)^2\right].$$

The noise covariances between output directions do not appear in R, so autocorrelation of the labels has no direct effect on the MSE risk.

Corollary 1 (Optimal weights)**.** *If the columns of β are independent with $\beta_{\cdot k} \sim (0, \frac{\alpha_k^2}{d}I_d)$, then $\mathbb{E}R_k(\lambda_k)$ is minimized at $\lambda_k^* = d\tilde{\sigma}_k^2/(n\alpha_k^2)$ for every spectrum s. With a single λ, the optimal weights are $\omega_k^* \propto \alpha_k^2/\tilde{\sigma}_k^2$, the signal-to-noise ratio (SNR) of direction k.*

Uniform weights, that is the MSE, are therefore optimal if and only if all directions have the same SNR. The GLS weights $\Omega = \Sigma_\varepsilon^{-1}$ that motivate QDF are optimal if and only if the signal is isotropic across the eigen-directions of Σ_ε; when signal and noise concentrate in the same (low-frequency) directions, GLS weights point the wrong way. Since $\lambda_k^* \to 0$ as $n/d \to \infty$, the gain from any weighting also vanishes on long series with small heads. Appendix A gives the exact risk for i.i.d. Gaussian features in the proportional limit; for sliding windows we have no such result.

Constrained heads. For a rank-r linear head, Ω changes the learned subspace. If the rank constraint is correct, every Ω gives a consistent estimator and $\Omega = \Sigma_\varepsilon^{-1}$ is asymptotically efficient (Anderson, 1951; Reinsel & Velu, 1998). If the rank is too small, Proposition 1(b) implies that weighting cannot reduce the population MSE, and it increases the MSE whenever it changes the minimizer (Proposition 5 in Appendix A). For a learned representation ϕ_θ with a linear head, the minimization over the head does not depend on Ω (Proposition 2), so Ω affects only which representation is learned, and we have no result on when this helps.

3.3. What Remains: The L1 Norm

By Proposition 3, under symmetric noise, L1-type losses aim at the same target as the MSE, and in finite samples they act as M-estimators whose influence is bounded in the residual, though not in the design (Huber, 1964). Two consequences of the analysis can be checked directly. L2 losses in any orthonormal basis should behave as the MSE, with the rFFT half-spectrum L2 differing only through its DC and Nyquist weights, and output-direction weighting should make little difference on unconstrained heads with $n \gg d$. A bounded influence can help in two different ways. In the classical one, heavy-tailed noise makes least squares inefficient, and the L1 fit can estimate the conditional mean more efficiently; with Gaussian noise, it is less efficient than least squares (Appendix B). In the other, a few training windows that are unlike the test period dominate the squared loss, and bounding their influence lowers the test error although it worsens the fit to the training windows. This second way needs a shift between training and test data and predicts no gain on stationary data. Because the bound acts on the residual, it limits windows with large errors, not windows with an unusual lookback. Which windows these are was not part of our initial analysis; the role of level shifts and instance normalization emerged from the experiments in Section 5.3, which point to the second way.

4. Experimental Protocol

Models and data. We run the official code at pinned commits. For iTransformer (Liu et al., 2024) and DLinear (Zeng et al., 2023), we use the FreDF repository with its scripts and hyper-parameters and a lookback of $L = 96$; for Time-o1, we use the code of its first release with its own scripts, which train Fredformer (Piao et al., 2024) on ETT and FreTS (Yi et al., 2023) on Weather. Our patches only add options, namely rolling borders, loss variants and normalization switches. Appendix E lists the commits and patches and compares our reproduction with the published numbers, which we report separately from our own results. Under its own protocol ($L = 336$), DLinear exceeds its published numbers by 5.0–5.9% on three datasets, for rea-

sons we could not find; our DLinear runs use the FreDF repository with $L = 96$ instead. We use ETTh1, ETTh2, ETTm1 and ETTm2 (Zhou et al., 2021) and Weather (Wu et al., 2021) with horizons $H \in \{96, 192, 336, 720\}$ and the standard 12/4/4-month splits for ETT. For rolling origins, we shift all ETT borders by $k \times 30$ days, $k = 0, \ldots, 4$, so that the test periods together cover 2017-10 to 2018-06. Because the origins overlap, they are not independent replicates. Weather is used only on the standard split, in the decomposition (Section 5.1), the per-channel analysis (Section 5.3), the weighting experiment (Section 5.4) and the linear models of Appendix C. The training runs took about 450 GPU job-hours, and reproducing the published numbers about 150 more (Appendix E).

Losses and selection. The main experiments compare MSE, FreDF with its official mixture, frequency L1, time L1 and time Huber. For the decomposition, we add full-spectrum FFT L2, rFFT L2, window $L_{2,1}$, PCA-L2, and PCA-L1 without Time-o1's standardization. Early stopping and all model choices use the validation MSE only; the test set is never used for selection. For equal-budget tuning on ETTh with iTransformer, we give each loss the same four configurations, learning rate $\times\{1, \frac{1}{4}\}$ combined with a halving or a constant schedule, select one by the validation MSE of one seed, and re-run the selected configuration with the other seeds. We use three seeds, and five for iTransformer on ETTh.

Evaluation. We report paired relative changes of test MSE, averaged over horizons and seeds, and the same comparisons in MAE in Appendix G. For each origin and horizon, we run a Diebold–Mariano test on the per-window errors, using the equal-weighted cosine long-run variance with $B = \lfloor T/2H \rfloor$ frequencies and t_B critical values (Lazarus et al., 2018; Coroneo & Iacone, 2020). In simulations with overlapping windows, this test has a size of 5–7.5% at the nominal 5% level, whereas a Newey–West estimator with bandwidth H rejects in 14–35% of the replications (Appendix F). We apply Holm's correction (Holm, 1979) within each family of 20 origin $\times$ horizon cells. As a reference, we compute for each dataset and origin the exact MSE solution of a channel-shared linear model on the lookback-mean-centred window, using ridge regression with λ chosen on validation data; with $n \gg d$ this choice changes its test MSE by at most 0.4% relative to $\lambda = 0$. Because it is the MSE optimum of its model class, it shows how far the SGD-trained models are from that optimum.

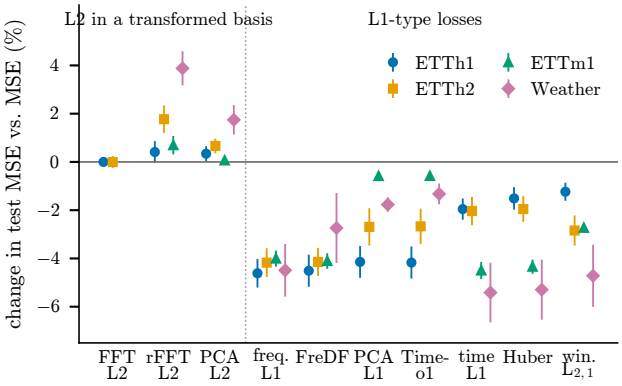


Figure 2. Decomposition of FreDF and Time-o1 (standard split). Paired change of test MSE relative to the same script trained with MSE (mean ± SEM over 4 horizons × 3 seeds). PCA-L2, PCA-L1 and Time-o1 are relative to the Time-o1 script (Fredformer on ETT, FreTS on Weather), the others to the FreDF script (iTransformer); FFT-L2 was run on ETTh1/ETTh2 only. Colour and marker shape encode the dataset; win. $L_{2,1}$ is the window $L_{2,1}$ loss.

5. Results

5.1. L2 and L1 Versions of FreDF and Time-o1

Figure 2 decomposes FreDF and Time-o1 on the standard split. L2 losses in a transformed basis do not improve on the MSE. The full-spectrum FFT L2 coincides with the MSE: per horizon, the means differ by at most 0.004% on ETTh1 and 0.2% on ETTh2, where single runs differ by up to 2.1%. PCA-L2 changes the test MSE by $+0.1\%$ to $+1.7\%$, and the rFFT half-spectrum L2 used in the FreDF code is worse by $+0.4\%$ to $+3.9\%$. The population analysis does not predict this degradation, since the minimizer is unchanged. Most of the extra error lies along the DC direction of the horizon, the mean level of the forecast, which the half spectrum weights twice (on ETTh2, $+1.6$ of $+1.8$ percentage points; nothing along the Nyquist direction; Table 6). In the linear model class, doubling only the DC weight raises the test MSE by 0.3–1.0%, and doubling only the Nyquist weight leaves it unchanged (Table 7). Replacing the L2 norm by an L1 norm recovers the full gain. Frequency L1 alone matches FreDF, with changes of -4.0% to -4.6% on all four datasets against -4.1% to -4.5% for FreDF on ETT and -2.7% on Weather, where FreDF's official mixture gives the MSE term a larger weight ($\lambda = 0.4$, against 0.1–0.2 on ETT). Likewise, PCA-L1 without Time-o1's standardization matches Time-o1, for example -4.1% versus -4.2% on ETTh1. Robust losses in the time domain also help. On ETTm1 and Weather they match or exceed the frequency L1 (-4.5% and -5.4% for time L1), while on ETTh1 and ETTh2 they reach about half of its gain. On eight consecutive sub-periods of the validation and test data, the L1 variants improve on the MSE in most, the L2 variants in few (Appendix G).

5.2. Rolling Origins: Stable L1 Gains, Backbone-Dependent Basis

Table 2 extends the comparison to five origins and two backbones. On iTransformer, all four L1-type losses improve on the MSE at every origin of every dataset, and no origin $\times$ horizon cell is significantly worse. FreDF reduces the test MSE by 3.0%, 3.7%, 2.0% and 5.4% on ETTh1, ETTh2, ETTm1 and ETTm2. Significant cells are concentrated on ETTm2 and ETTh, since the test has little power at long horizons ($B = 1$ at $H = 720$). Single training runs agree: on iTransformer, each L1-type loss beats the MSE of the same seed in 87–100% of the origin $\times$ horizon $\times$ seed cells. The gains are of similar size in MAE (Table 4), so they are not only a closer match between the training loss and the metric. Tuning the MSE with the same validation-based budget does not close the gap. Tuned FreDF beats tuned MSE by 2.9% and 2.8% on ETTh1 and ETTh2, at every origin, and tuning barely changes the test error of the MSE itself. On the standard split, a grid that also varies the dropout never selects the official configuration; tuned FreDF still beats tuned MSE by 3.3–3.7% on ETT but by only 0.3% on Weather, where the untuned time and frequency L1 losses beat the tuned MSE by 3.3% and 2.3% (Appendix G). The basis, in contrast, does not give a consistent advantage. Frequency L1 is 0.1–1.5% better than time L1 on iTransformer, but 0.8–4.4% worse on DLinear at every dataset and origin. In the 20 dataset $\times$ origin combinations, MSE-trained iTransformer is 0–8% worse than the exact MSE solution of the linear reference, and FreDF is within -4.2% to $+3.6\%$ of it (Appendix G).

5.3. Mechanism: Instance Normalization and the Largest Training Errors

Early stopping. Frequency L1 delays the best validation epoch of iTransformer, from about 1.5 to 4–7 epochs on ETTh, but this does not explain its gain. Against the MSE stopped at its test-optimal epoch, an oracle choice, FreDF still leads by 1.2–4.8% and time L1 by 1.0–3.6% on the four ETT datasets, in 73–100% of the cells. Time L1 gains without delaying the best epoch, and the change in the best epoch is unrelated to the change in test MSE (Spearman ρ between -0.3 and 0.5).

Instance normalization. Table 2 shows much larger L1 gains for DLinear than for iTransformer on ETTh2 and ETTm2. Because DLinear does not normalize its inputs, we then ran controls that vary normalization while keeping each model otherwise fixed (Figure 3). In the linear model, removing the centring multiplies the gain of L1-type losses by 1.7 to 16, most on ETTh2 and ETTm2. Adding RevIN to DLinear almost removes its large gains; for time L1, the change relative to the MSE shrinks from -14.3% to -0.5% on ETTh2 and from -9.8% to -0.2% on ETTm2. DLinear with RevIN trained by the MSE also reaches the level of DLinear without normalization trained by L1, with test MSE 0.386 versus 0.388 on ETTh2 and 0.262 versus 0.259 on ETTm2. Conversely, switching off the built-in normalization of iTransformer increases the L1 gain on ETTh2 and ETTm2 to 13–18%, although the unnormalized model remains far worse than the normalized one.

The per-window errors show where the MSE loses without normalization. On ETTh2 and ETTm2, MSE-trained DLinear is better than its L1 counterpart on the 10% of windows with the largest level offset (-11.6% and -22.4%) but worse on the remaining 90% ($+22.5\%$ and $+19.1\%$). The training windows do not show this trade-off (Table 5). There, the L1 fit is worse than the MSE fit on the windows with the largest offset ($+16\%$ and $+21\%$) and about equal on the others. The training error is also about as concentrated with RevIN, where L1 does not help: in both cases the 10% of windows with the largest error carry about half of it. The L1 advantage therefore appears out of sample, where the test level lies about one standard deviation away from the training level on these datasets, and not as a better fit of the bulk of the training windows. Across the 20 dataset $\times$ origin cells, the time-L1 gain of DLinear grows with the level offset of the test period (Spearman $\rho = -0.88$ between offset and change of test MSE); on ETTh2, the change goes from -22.3% at origin 0, with offset 1.29, to -8.4% at origin 4, with offset 0.60. iTransformer with its normalization switched off shows the same relation ($\rho = -0.82$), whereas with normalization the gain does not grow with the offset in any dataset (Appendix H). The origins overlap and differ in season, and the four datasets come from two transformers, so this check stays within the ETT family. Interventions on the training loss of the DLinear-like linear model single out the same windows (Table 7). Dropping the 10% of windows with the largest error in each batch recovers 66% of the time-L1 gain on ETTh2 and all of it on ETTm2 (-7.2% and -6.8%), and 42% and 68% of the small gains on ETTh1 and ETTm1. Dropping the 10% with the largest level offset recovers 56% and all of it on ETTh2 and ETTm2 (-6.2% and -6.5%), whereas clipping the global gradient norm recovers little (-1.0% and -1.8%). Rescaling the squared error of each horizon step and channel by a running mean, which balances magnitudes without bounding single windows, recovers 44% and 61% (-4.9% and -3.8%). Without normalization, the L1 losses thus help by keeping the largest training errors, concentrated in a few windows and channels, from dominating a fit that carries over badly to a drifted test period. This is robust estimation in the second sense of Section 3.3: the L1 fit is worse on the training windows, and its gain depends on the shift.

With normalization, L1-type losses still improve iTransformer at every origin, by 1.3–5.5% on average over origins

Table 2. Rolling-origin evaluation (4 ETT datasets × 5 origins × 4 horizons). Each cell: paired relative change of test MSE (%, negative = better), mean over origins of the horizon × seed means; in parentheses the number of origins with a negative mean; then the number of origin × horizon cells (of 20) where the comparison is significantly better/worse in a Diebold–Mariano test (EWC long-run variance, Holm-corrected within each row and dataset). The origins are 30 days apart with 4-month test periods, so they overlap and the counts are descriptive, not independent replicates. Official hyper-parameters for every loss; tuned rows: each loss tuned on validation data with the same grid.

Backbone	Comparison	ETTh1	ETTh2	ETTm1	ETTm2
iTransformer	FreDF vs. MSE	−3.0 (5/5) 6/0	−3.7 (5/5) 7/0	−2.0 (5/5) 2/0	−5.4 (5/5) 14/0
iTransformer	frequency L1 vs. MSE	−3.0 (5/5) 6/0	−3.8 (5/5) 6/0	−2.0 (5/5) 2/0	−5.5 (5/5) 14/0
iTransformer	time L1 vs. MSE	−1.5 (5/5) 0/0	−2.4 (5/5) 2/0	−1.9 (5/5) 2/0	−4.2 (5/5) 11/0
iTransformer	time Huber vs. MSE	−1.3 (5/5) 0/0	−2.4 (5/5) 3/0	−1.7 (5/5) 3/0	−4.1 (5/5) 12/0
iTransformer	frequency vs. time L1	−1.5 (5/5) 1/0	−1.4 (5/5) 0/0	−0.1 (3/5) 0/1	−1.4 (5/5) 1/0
DLinear	frequency L1 vs. MSE	+1.9 (0/5) 0/0	−12.9 (5/5) 2/0	+0.5 (1/5) 0/1	−9.0 (5/5) 7/0
DLinear	time L1 vs. MSE	−2.4 (5/5) 4/0	−14.3 (5/5) 3/0	−1.0 (4/5) 0/0	−9.8 (5/5) 8/0
DLinear	time Huber vs. MSE	−2.4 (5/5) 4/0	−13.8 (5/5) 3/0	−1.3 (5/5) 0/0	−10.3 (5/5) 12/0
DLinear	frequency vs. time L1	+4.4 (0/5) 0/3	+1.5 (0/5) 0/0	+1.5 (0/5) 0/1	+0.8 (0/5) 0/2
iTransformer	tuned FreDF vs. tuned MSE	−2.9 (5/5)	−2.8 (5/5)	–	–
iTransformer	tuned time L1 vs. tuned MSE	−1.5 (5/5)	−2.2 (5/5)	–	–

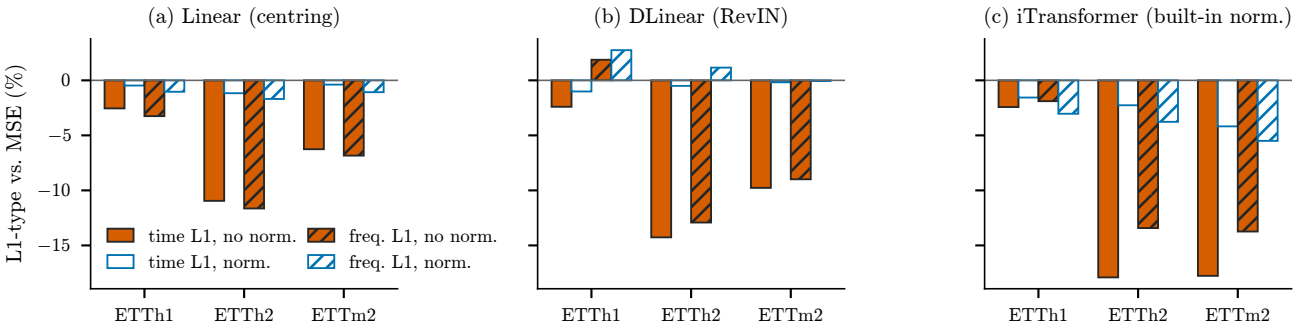


Figure 3. Gain of L1-type losses over MSE (%, test, mean over 5 origins × 4 horizons × seeds) with and without instance normalization. (a) The same linear model with or without lookback-mean centring (Adam, official protocol). (b) DLinear, official (no normalization) vs. RevIN. (c) iTransformer, official built-in normalization vs. switched off. Filled bars: without normalization; open bars: with normalization; hatching: frequency L1. Panel (c) uses the three seeds of the normalization-off runs, so on ETTh2 its value with normalization (−2.3%) differs from Table 2, which uses five seeds (−2.4%).

(Table 2), and the gain remains after tuning and against oracle stopping, so normalization does not account for it. On the standard split, 57–62% of the frequency-L1 gain lies along the DC direction (Table 6), so most of the remaining gain is a smaller error in the forecast level. In the centred linear model, the small frequency-L1 gain on ETTh1 and ETTh2 is reproduced by an L2 loss whose frequencies are balanced by a running scale (−1.0% and −1.7%; Table 7). On the standard split, the official protocol leaves that model's MSE fit 1.6% and 2.2% above the exact solution on these datasets (Appendix C), so this gain is one of optimization, in line with the gradient-shape account of An et al. (2026).

Weather. Time L1 is 5.9% worse than the exact MSE solution in the centred linear model (Appendix C) but gives iTransformer its largest gain (Section 5.1). The linear model loses on the skewed precipitation channels and on wind (+7% to +14%), whereas iTransformer gains on precipitation too (−3.7% and −4.9%) and most on pressure (−17%) and radiation (−21% to −23%). Why iTransformer escapes the bias on the skewed channels, we cannot say.

5.4. Output-Direction Weighting Has Almost No Effect on Linear Heads

We fit the linear reference in closed form on 168 settings (4 ETT datasets × 5 origins × $L \in \{96, 336\}$ × 4 horizons, plus Weather) and compare uniform ridge with several weighting schemes. The schemes are per-direction ridge parameters chosen on validation data (in time-step, Fourier or residual-PCA directions), GLS weights $\hat{\Sigma}_\varepsilon^{-1}$, label-PCA truncation, and the SNR plug-in of Corollary 1. Relative to uniform ridge, per-direction tuning changes the test MSE by less than 0.1% on average, although it improves the validation MSE by 0.1–1.2%. GLS weights change the test MSE by +0.1% on average and are better in

40% of the settings, and an orthogonal rotation alone gives bit-identical results. With $n \approx 5.3 \times 10^4$ to 7.7×10^5 rows and $d \leq 335$, this is what Corollary 1 predicts. A plug-in estimate of the SNR weights is worse than uniform ridge (Appendix I). Such heads are, however, where the theory predicts the least effect, and QDF's learned Ω in a deep model is not tested here.

5.5. Selecting the Loss on Validation Data

Choosing the loss with the lowest validation MSE works on iTransformer, where the chosen loss beats the MSE in 80–100% of the origin $\times$ horizon cells, but not on DLinear for ETTh1 and ETTm1, where it does so in 40% and 60%. The failure follows the season: validation periods in summer and autumn favour the MSE, whereas the winter test periods favour L1. Neither of two remedies fixes this. Pooling all earlier out-of-sample windows lowers the two win rates to 30% and 50%. On a DLinear-like linear model, a validation block from the same season one year earlier fixes ETTh1 but not ETTm1, and it breaks ETTh2 and ETTm2, because the block lies inside the training span and does not show the forward level drift behind most of the L1 gain there. Always using the time Huber loss beats the MSE in 90–100% of the DLinear cells; since this comparison uses test results, it can only justify a default choice (Appendix D).

6. Discussion

Practical implications. The first decision is the normalization rather than the loss. Once each input window is normalized or centred, robust losses still give modest gains on iTransformer and small or no gains on DLinear; without normalization, a robust loss such as time Huber is the safer default. A single validation period is a poor guide for choosing between the MSE and a robust loss, because in our data neither the preceding months nor the same season of the previous year reflected both seasonality and level drift; validation data still serve for early stopping. Finally, conclusions about the basis can depend on the months of a single test period (Appendix G), which argues for rolling origins and Diebold–Mariano tests with fixed-smoothing variance estimates.

Limitations. Our evidence covers the ETT family under rolling origins and Weather on the standard split, with two backbones, iTransformer and DLinear, plus Fredformer and FreTS in the Time-o1 decomposition, all with a lookback of $L = 96$; we did not run electricity or traffic data or other architectures. The four ETT datasets come from two transformers with marked level drift, so how strongly the L1 gain depends on normalization may partly be a property of this family. The interventions on the training loss ran only in the linear model class, so whether they reproduce the remaining gain of normalized iTransformer is open. QDF enters only through the theory, the linear weighting experiment and our reproduction, so its mechanism on deep models is untested. The tuning grid covers the learning rate, the schedule and, on the standard split, the dropout, but not FreDF's mixing weight. The risk formula of Proposition 4 does not cover sliding windows, and the seasonal-validation experiment uses a linear model with two seeds and a single previous year. Finally, normalization does not explain the average gain of 1.3–5.5% that L1-type losses still achieve on iTransformer, a large part of which lies in the forecast level. Why an MSE-trained iTransformer stays above the exact MSE solution of a linear model, and why L1 closes this gap, remain open questions.

7. Conclusion

Horizon-aware losses do not correct a decorrelation bias of the MSE: in any orthonormal basis, their L2 versions share the minimizer of the MSE and match or trail it in practice. Their gains come from the L1 norm. Without instance normalization, it bounds the influence of the largest training errors, which otherwise dominate a fit that transfers badly to a drifted test period; with normalization, a smaller gain remains, mostly in the forecast level, and its cause is open. Inputs should thus be normalized before a loss is chosen.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

Ali, A., Kolter, J. Z., and Tibshirani, R. J. A Continuous-Time View of Early Stopping for Least Squares Regression. In *International Conference on Artificial Intelligence and Statistics (PMLR 89)*, 2019. URL https://proceedings.mlr.press/v89/ali19a.html.

An, Z., You, J., Li, J., Tang, Y., Li, W., Du, H., and Du, S. FreDN: Spectral Disentanglement for Time Series Forecasting via Learnable Frequency Decomposition. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(24):19623–19631, 2026. doi: 10.1609/aaai.v40i24.39042.

Anderson, T. W. Estimating Linear Restrictions on Regression Coefficients for Multivariate Normal Distributions. *The Annals of Mathematical Statistics*, 22(3):327–351, 1951. doi: 10.1214/aoms/1177729580.

Ben Taieb, S. and Atiya, A. F. A Bias and Variance Analysis for Multistep-Ahead Time Series Forecasting. *IEEE Transactions on Neural Networks and Learning Systems*, 27(1):62–76, 2016. doi: 10.1109/tnnls.2015.2411629.

Ben Taieb, S., Bontempi, G., Atiya, A. F., and Sorjamaa, A. A review and comparison of strategies for multi-step ahead time series forecasting based on the NN5 forecasting competition. *Expert Systems with Applications*, 39 (8):7067–7083, 2012. doi: 10.1016/j.eswa.2012.01.039.

Bergmeir, C. and Benítez, J. M. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012. doi: 10.1016/j.ins.2011.12.028.

Berthelier, G., Nabil, T., Naour, E. L., Niamke, R., Perlaza, S., and Neglia, G. On the Role of Reversible Instance Normalization. arXiv:2603.11869, 2026. URL https://arxiv.org/abs/2603.11869.

Breiman, L. and Friedman, J. H. Predicting Multivariate Responses in Multiple Linear Regression. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 59(1):3–54, 1997. doi: 10.1111/1467-9868.00054.

Cai, R., Wan, Y., Zhang, K., Jin, M., Ge, Z., Dong, D., Yu, H., Liu, Y., and Wen, Q. The Procrustean Bed of Time Series: The Optimization Bias in Point-wise Loss Functions. arXiv:2512.18610, 2025. URL https://arxiv.org/abs/2512.18610.

Chevillon, G. Direct Multi-Step Estimation and Forecasting. *Journal of Economic Surveys*, 21(4):746–785, 2007. doi: 10.1111/j.1467-6419.2007.00518.x.

Coroneo, L. and Iacone, F. Comparing predictive accuracy in small samples using fixed-smoothing asymptotics. *Journal of Applied Econometrics*, 35(4):391–409, 2020. doi: 10.1002/jae.2756.

Diebold, F. X. and Mariano, R. S. Comparing Predictive Accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.

Dobriban, E. and Wager, S. High-dimensional asymptotics of prediction: Ridge regression and classification. *The Annals of Statistics*, 46(1), 2018. doi: 10.1214/17-aos1549.

Hastie, T., Montanari, A., Rosset, S., and Tibshirani, R. J. Surprises in high-dimensional ridgeless least squares interpolation. *The Annals of Statistics*, 50(2), 2022. doi: 10.1214/21-aos2133.

Hewamalage, H., Ackermann, K., and Bergmeir, C. Forecast evaluation for data scientists: common pitfalls and best practices. *Data Mining and Knowledge Discovery*, 37(2):788–832, 2022. doi: 10.1007/s10618-022-00894-5.

Holm, S. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6:65–70, 1979. doi: 10.2307/4615733.

Hounie, I., Porras-Valenzuela, J., and Ribeiro, A. Loss Shaping Constraints for Long-Term Time Series Forecasting. In *International Conference on Machine Learning (PMLR 235)*, 2024. URL https://proceedings.mlr.press/v235/hounie24a.html.

Huber, P. J. Robust Estimation of a Location Parameter. *The Annals of Mathematical Statistics*, 35(1):73–101, 1964. doi: 10.1214/aoms/1177703732.

Izenman, A. J. Reduced-rank regression for the multivariate linear model. *Journal of Multivariate Analysis*, 5(2):248–264, 1975. doi: 10.1016/0047-259x(75)90042-1.

Kim, T., Kim, J., Tae, Y., Park, C., Choi, J.-H., and Choo, J. Reversible Instance Normalization for Accurate Time-Series Forecasting against Distribution Shift. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=cGDAkQo1C0p.

Kruskal, W. When are Gauss-Markov and Least Squares Estimators Identical? A Coordinate-Free Approach. *The Annals of Mathematical Statistics*, 39(1):70–75, 1968. doi: 10.1214/aoms/1177698505.

Kudrat, D., Xie, Z., Sun, Y., Jia, T., and Hu, Q. Patch-wise Structural Loss for Time Series Forecasting. In *International Conference on Machine Learning (PMLR 267)*, 2025. URL https://proceedings.mlr.press/v267/kudrat25a.html.

Lazarus, E., Lewis, D. J., Stock, J. H., and Watson, M. W. HAR Inference: Recommendations for Practice. *Journal of Business & Economic Statistics*, 36(4):541–559, 2018. doi: 10.1080/07350015.2018.1506926.

Le Guen, V. and Thome, N. Shape and Time Distortion Loss for Training Deep Time Series Forecasting Models. In *Advances in Neural Information Processing Systems*, 2019. URL https://papers.nips.cc/paper_files/paper/2019/hash/466accbac9a66b805ba50e42ad715740-Abstract.html.

Li, Z., Qi, S., Li, Y., and Xu, Z. Revisiting Long-term Time Series Forecasting: An Investigation on Linear Mapping. arXiv:2305.10721, 2023. URL https://arxiv.org/abs/2305.10721.

Liu, H., Aue, A., and Paul, D. On the marčenko–pastur law for linear time series. *The Annals of Statistics*, 43(2), 2015. doi: 10.1214/14-aos1294.

Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L., and Long, M. iTransformer: Inverted Transformers Are Effective for Time Series Forecasting. In *International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=JePfAI8fah.

Nie, Y., Nguyen, N. H., Sinthong, P., and Kalagnanam, J. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=Jbdc0vTOcol.

Piao, X., Chen, Z., Murayama, T., Matsubara, Y., and Sakurai, Y. Fredformer: Frequency Debiased Transformer for Time Series Forecasting. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 2400–2410, 2024. doi: 10.1145/3637528.3671928.

Qiu, X., Hu, J., Zhou, L., Wu, X., Du, J., Zhang, B., Guo, C., Zhou, A., Jensen, C. S., Sheng, Z., and Yang, B. TFB: Towards Comprehensive and Fair Benchmarking of Time Series Forecasting Methods. *Proceedings of the VLDB Endowment*, 17(9):2363–2377, 2024. doi: 10.14778/3665844.3665863.

Qiu, X., Wu, X., Cheng, H., Liu, X., Guo, C., Hu, J., and Yang, B. DBLoss: Decomposition-based Loss Function for Time Series Forecasting. In *Advances in Neural Information Processing Systems*, 2025. URL https://papers.nips.cc/paper_files/paper/2025/hash/2823c4f838727564c108c85d7e6e8507-Abstract-Conference.html.

Reinsel, G. C. and Velu, R. P. *Multivariate Reduced-Rank Regression*. Springer, 1998. doi: 10.1007/978-1-4757-2853-8.

Shi, T., Meng, Z., Chen, Y., Zheng, S., Su, F., Huang, J., Ren, C., and Zhao, Z. OLMA: One Loss for More Accurate Time Series Forecasting. arXiv:2505.11567, 2025. URL https://arxiv.org/abs/2505.11567.

Tashman, L. J. Out-of-sample tests of forecasting accuracy: an analysis and review. *International Journal of Forecasting*, 16(4):437–450, 2000. doi: 10.1016/s0169-2070(00)00065-0.

Toner, W. and Darlow, L. N. An Analysis of Linear Time Series Forecasting Models. In *International Conference on Machine Learning (PMLR 235)*, 2024. URL https://proceedings.mlr.press/v235/toner24a.html.

Wang, H., Pan, L., Chen, Z., Chen, X., Dai, Q., Wang, L., Li, H., and Lin, Z. Time-o1: Time-Series Forecasting Needs Transformed Label Alignment. In *Advances in Neural Information Processing Systems*, 2025a. URL https://papers.nips.cc/paper_files/paper/2025/hash/0cd62dea69635f4c5b569848267fe5a8-Abstract-Conference.html.

Wang, H., Pan, L., Chen, Z., Yang, D., Zhang, S., Yang, Y., Liu, X., Li, H., and Tao, D. FreDF: Learning to Forecast in the Frequency Domain. In *International Conference on Learning Representations*, 2025b. URL https://openreview.net/forum?id=4A9IdSa1ul.

Wang, H., Pan, L., Lu, Y., Chen, Z., Liu, T., He, S., Chu, Z., Wen, Q., Li, H., and Lin, Z. Quadratic Direct Forecast for Training Multi-Step Time-Series Forecast Models. In *International Conference on Learning Representations*, 2026a. URL https://openreview.net/forum?id=vpO8n9AqEG.

Wang, H., Pan, L., Lu, Y., Chu, Z., Li, X., He, S., Chen, Z., Li, H., Wen, Q., and Lin, Z. DistDF: Time-series Forecasting Needs Joint-distribution Wasserstein Alignment. In *International Conference on Learning Representations*, 2026b. URL https://openreview.net/forum?id=VrdLwUmzBy.

Wang, H., Pan, L., Wen, Q., Yu, J., Chen, Z., Zheng, C., Li, X., Chu, Z., Xu, C., Gong, M., Li, H., Lu, Y., Lin, Z., Torr, P., and Liu, Y. Deep Autocorrelation Modeling for Time-Series Forecasting: Progress and Prospects. arXiv:2603.19899, 2026c. URL https://arxiv.org/abs/2603.19899.

Wang, J., Shang, X., Li, F., and Peng, F. RI-Loss: A Learnable Residual-Informed Loss for Time Series Forecasting. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(31):26277–26284, 2026d. doi: 10.1609/aaai.v40i31.39832.

Wu, D. and Xu, J. On the Optimal Weighted ℓ_2 Regularization in Overparameterized Linear Regression. In *Advances in Neural Information Processing Systems*, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/72e6d3238361fe70f22fb0ac624a7072-Abstract.html.

Wu, H., Xu, J., Wang, J., and Long, M. Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting. In *Advances in Neural Information Processing Systems*, 2021. URL https://papers.nips.cc/paper_files/paper/2021/hash/bcc0d400288793e8bdcd7c19a8ac0c2b-Abstract.html.

Xu, Z., Zeng, A., and Xu, Q. FITS: Modeling Time Series with 10k Parameters. In *International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=bWcnvZ3qMb.

Yi, K., Zhang, Q., Fan, W., Wang, S., Wang, P., He, H., An, N., Lian, D., Cao, L., and Niu, Z. Frequency-domain MLPs are More Effective Learners in Time Series Forecasting. In *Advances in Neural Information Processing Systems*, 2023. URL https://papers.nips.cc/paper_files/paper/2023/hash/f1d16af76939f476b5f040fd1398c0a3-Abstract-Conference.html.

Zellner, A. An Efficient Method of Estimating Seemingly Unrelated Regressions and Tests for Aggregation Bias. *Journal of the American Statistical Association*, 57(298): 348–368, 1962. doi: 10.1080/01621459.1962.10480664.

Zeng, A., Chen, M., Zhang, L., and Xu, Q. Are Transformers Effective for Time Series Forecasting? *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(9): 11121–11128, 2023. doi: 10.1609/aaai.v37i9.26317.

Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., and Zhang, W. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(12):11106–11115, 2021. doi: 10.1609/aaai.v35i12.17325.

A. Proofs

Unit tests in the released code check the identities of this appendix numerically: the rFFT weights, Theorem 1(a)–(c), the full-rank case of Proposition 5 and the Marchenko–Pastur transform of Proposition 4.

A.1. The rFFT Half-Spectrum L2 (Table 1)

Let $e \in \mathbb{R}^H$ be real, H even, $F_k = \sum_{t=0}^{H-1} e_t \, \mathrm{e}^{-2\pi \mathrm{i} t k/H}$, and $K = H/2 + 1$. By Parseval, $\sum_{k=0}^{H-1} |F_k|^2 = H\|e\|^2$, and $|F_k| = |F_{H-k}|$ for real e. Hence

$$\sum_{k=0}^{H/2} |F_k|^2 = \tfrac{1}{2}\big(H\|e\|^2 + |F_0|^2 + |F_{H/2}|^2\big) = \tfrac{1}{2}\, e^\top \big(HI + \mathbf{1}\mathbf{1}^\top + aa^\top\big) e,$$

since $F_0 = \mathbf{1}^\top e$ and $F_{H/2} = a^\top e$ with $a_t = (-1)^t$. Dividing by K gives $\Omega = \frac{H}{2K}\big(I + \frac{1}{H}(\mathbf{1}\mathbf{1}^\top + aa^\top)\big)$. Its eigenvalue is H/K on $\mathbf{1}$ and a and $H/(2K)$ on their orthogonal complement, so the mean and the Nyquist directions of the error are weighted twice. For odd H, the Nyquist term is absent.

A.2. Proposition 1

Write $y - f(x) = \varepsilon + (m(x) - f(x))$. The cross term vanishes, since $\mathbb{E}[\varepsilon^\top \Omega (m-f)(x)] = \mathbb{E}\big[\mathbb{E}[\varepsilon \mid x]^\top \Omega (m-f)(x)\big] = 0$. This gives the decomposition, and part (a) follows because the second term is non-negative and zero at $f = m$. Part (b) holds by definition of the minimum, and the inequality is strict whenever f_Ω is not itself an MSE minimizer. If $\Omega \succeq 0$ is singular, the decomposition still holds, the minimizer is determined only on $\mathrm{range}(\Omega)$, and the components in $\ker \Omega$ are fixed by the initialization or the architecture. □

A.3. Proposition 2

Let $r = m - \Pi_{\mathcal{G}} m$, whose coordinates are orthogonal to $\mathcal{G}$ in $L^2(P_x)$. For $f \in \mathcal{F}$,

$$\mathbb{E}\|m - f\|_\Omega^2 = \mathbb{E}\|r\|_\Omega^2 + \mathbb{E}\|\Pi_{\mathcal{G}} m - f\|_\Omega^2 + 2 \sum_{h,h'} \Omega_{hh'} \mathbb{E}\big[r_h \, (\Pi_{\mathcal{G}} m - f)_{h'}\big],$$

and the last term is zero because $(\Pi_{\mathcal{G}} m - f)_{h'} \in \mathcal{G}$. The middle term is minimized, uniquely in L^2 since $\Omega \succ 0$, by $f = \Pi_{\mathcal{G}} m$, for every Ω. Together with Proposition 1, this proves the claim. □

A.4. Proposition 3

Assume $\mathbb{E}[\|\varepsilon\| \mid x] < \infty$. Fix x and let $g(\mu) = \mathbb{E}[\psi(y - \mu) \mid x]$, which is convex because ψ is a sum of norms of linear maps. For any δ, $g(m + \delta) = \mathbb{E}[\psi(\varepsilon - \delta) \mid x]$ and $g(m - \delta) = \mathbb{E}[\psi(\varepsilon + \delta) \mid x] = \mathbb{E}[\psi(-\varepsilon + \delta) \mid x] = \mathbb{E}[\psi(\varepsilon - \delta) \mid x]$, using the symmetry of $\varepsilon \mid x$ and $\psi(-u) = \psi(u)$. By convexity, $g(m) \le \frac{1}{2} g(m + \delta) + \frac{1}{2} g(m - \delta) = g(m + \delta)$. □

A.5. Theorem 1

(a) Write $J(B) = \|(Y - \Phi B)\Omega^{1/2}\|_F^2$ and $C = B\Omega^{1/2}$. Then $J = \|Y\Omega^{1/2} - \Phi C\|_F^2$, whose minimizers are $C = \Phi^+ Y \Omega^{1/2} + N$ with $\Phi N = 0$. Hence the minimizers in B are $\Phi^+ Y + N\Omega^{-1/2}$, that is, $\{\Phi^+ Y + N' : \Phi N' = 0\}$, which is the same set for every $\Omega \succ 0$. Its minimum-norm element is $\Phi^+ Y$, whose columns lie in the row space of Φ.

(b) The first-order condition is $SB\Omega + \lambda B = Z\Omega$ with $Z = \Phi^\top Y/n$. Multiplying by $U^\top$ on the left and V on the right gives $\mathrm{diag}(s)\tilde{B}\,\mathrm{diag}(\omega) + \lambda \tilde{B} = \tilde{Z}\,\mathrm{diag}(\omega)$ with $\tilde{Z} = U^\top Z V$, that is $(s_j \omega_k + \lambda)\tilde{B}_{jk} = \omega_k \tilde{Z}_{jk}$. For $s_j > 0$, $\tilde{B}_{jk}^{\mathrm{OLS}} = \tilde{Z}_{jk}/s_j$ gives the claim; for $s_j = 0$, $\tilde{Z}_{jk} = 0$ and $\tilde{B}_{jk} = 0$.

(c) Gradient descent on $\frac{1}{2n}\,\mathrm{tr}((Y - \Phi B)\Omega(Y - \Phi B)^\top)$ reads $B_{t+1} = B_t + \eta(Z - SB_t)\Omega$, i.e. $\tilde{B}_{t+1,jk} = \tilde{B}_{t,jk} + \eta\omega_k(\tilde{Z}_{jk} - s_j \tilde{B}_{t,jk})$. From $\tilde{B}_0 = 0$ this geometric recursion gives $\tilde{B}_{t,jk} = [1 - (1 - \eta s_j \omega_k)^t]\tilde{Z}_{jk}/s_j$. The iterates converge to $\Phi^+ Y$ when $\eta < 2/(\omega_{\max} s_{\max})$. □

Risk decomposition. Let $Y = \Phi B_* + E$ with a fixed design, independent rows of E with covariance Σ_ε, and define the excess risk $\frac{1}{n}\mathbb{E}\|\Phi(\hat{B} - B_*)\|_F^2 = \sum_{j,k} s_j \mathbb{E}(\hat{\tilde{B}}_{jk} - \beta_{jk})^2$. Since $u_j^\top \Phi^\top E v_k$ has variance $\|\Phi u_j\|^2 \tilde{\sigma}_k^2 = n s_j \tilde{\sigma}_k^2$, $\tilde{B}_{jk}^{\mathrm{OLS}}$ is

unbiased for β_{jk} with variance $\tilde{\sigma}_k^2/(ns_j)$. With $c_{jk} = s_j/(s_j + \lambda_k)$ and $\lambda_k = \lambda/\omega_k$, $\hat{\tilde{B}}_{jk} = c_{jk}\tilde{B}_{jk}^{\mathrm{OLS}}$ has mean squared error $(1 - c_{jk})^2\beta_{jk}^2 + c_{jk}^2\tilde{\sigma}_k^2/(ns_j)$. Multiplying by s_j and summing over j gives R_k; no covariance between different k enters.

A.6. Corollary 1

With $\mathbb{E}\beta_{jk}^2 = \alpha_k^2/d =: a$ and $b := \tilde{\sigma}_k^2/n$, each summand of $\mathbb{E}R_k$ is $f_j(\lambda) = (s_j\lambda^2 a + s_j^2 b)/(s_j + \lambda)^2$, and $f_j'(\lambda) = 2s_j^2(\lambda a - b)/(s_j + \lambda)^3$. Every summand is minimized at $\lambda = b/a = d\tilde{\sigma}_k^2/(n\alpha_k^2)$, whatever s_j, hence so is the sum. With a common λ, $\lambda_k = \lambda/\omega_k$ equals λ_k^* for all k if and only if $\omega_k \propto \alpha_k^2/\tilde{\sigma}_k^2$. For $\Omega = \Sigma_\varepsilon^{-1}$ in the eigenbasis of Σ_ε, $\omega_k = 1/\tilde{\sigma}_k^2$, which is proportional to the SNR if and only if all α_k^2 are equal. □

A.7. Proportional Asymptotics

Proposition 4. *Let the rows $\phi(x_i)$ be i.i.d. $N(0, I_d)$, $d/n \to \gamma \in (0, \infty)$ with H fixed, the noise rows independent of the design and of each other with covariance Σ_ε, and β as in Corollary 1 with fixed α_k^2. For fixed $\omega \succ 0$ and $\lambda > 0$, the out-of-sample excess risk of the estimator in Theorem 1(b) converges to $\sum_{k=1}^H r(\lambda/\omega_k; \alpha_k^2, \tilde{\sigma}_k^2, \gamma)$, where $r(\lambda; \alpha^2, \sigma^2, \gamma) = \lambda^2\alpha^2 m'(-\lambda) + \sigma^2\gamma\big(m(-\lambda) - \lambda m'(-\lambda)\big)$ and m is the Stieltjes transform of the Marchenko–Pastur law with ratio γ. Each term is minimized at $\lambda/\omega_k = \gamma\tilde{\sigma}_k^2/\alpha_k^2$.*

Proof sketch. By Theorem 1(b), the estimator decouples along the columns of V into H ridge regressions of Yv_k on Φ, with penalty λ/ω_k and noise $v_k^\top\varepsilon_i$ of variance $\tilde{\sigma}_k^2$, independent of the design. The limiting risk of each is given by Dobriban & Wager (2018, Thm. 2.1) (see also Hastie et al., 2022); summing over the fixed number H of directions gives the claim. In simulations, the expression matches Monte Carlo risks within 2% for $\gamma \in [0.025, 0.5]$ (Appendix B). □

Sliding windows (not covered). In forecasting, the rows are overlapping windows of one series, so neighbouring rows share $L - 1$ values and the H-step noise is correlated with the features of later rows. Marchenko–Pastur-type limits exist for linear time series (Liu et al., 2015), but we do not have a result for the ridge risk under lag-embedded designs with overlapping multi-step noise. Appendix B shows that the OLS risk is close to the i.i.d. case while the optimal penalty differs by a factor of about ten, and Appendix I shows that a plug-in based on the i.i.d. formulas fails on real data.

A.8. Constrained Heads

Proposition 5. *Let $\mathcal{F}_r = \{x \mapsto B^\top x : \operatorname{rank} B \le r\}$, $m(x) = B_*^\top x$ and $\Sigma_x = \mathbb{E}xx^\top \succ 0$. The minimizer of R_Ω over $\mathcal{F}_r$ is $B_{\Omega,r} = B_*\Omega^{1/2}P_r\Omega^{-1/2}$, where P_r projects onto the top-r eigenvectors of $\Omega^{1/2}B_*^\top\Sigma_x B_*\Omega^{1/2}$. If $\operatorname{rank} B_* \le r$, then $B_{\Omega,r} = B_*$ for every Ω; otherwise $\mathrm{MSE}(B_{\Omega,r}) \ge \mathrm{MSE}(B_{I,r})$, strictly unless $B_{\Omega,r}$ also minimizes the MSE over $\mathcal{F}_r$.*

Proof. We have $R_\Omega(B) - \mathbb{E}\|\varepsilon\|_\Omega^2 = \|\Sigma_x^{1/2}(B_* - B)\Omega^{1/2}\|_F^2$. With $C = \Sigma_x^{1/2}B_*\Omega^{1/2}$, the best rank-$r$ approximation of C is CP_r (Eckart–Young), with P_r the projection onto the top-r right singular vectors of C, the top eigenvectors of $C^\top C$. Solving $\Sigma_x^{1/2}B\Omega^{1/2} = CP_r$ gives $B_{\Omega,r}$. If $\operatorname{rank} B_* \le r$, $CP_r = C$. The last statement is Proposition 1(b). □

Under a correct rank, it is classical that $\Omega = \Sigma_\varepsilon^{-1}$ gives the Gaussian maximum-likelihood estimator, which is asymptotically efficient (Anderson, 1951; Izenman, 1975; Reinsel & Velu, 1998). For a learned representation with a linear head, $\mathcal{F} = \{x \mapsto W^\top\phi_\theta(x)\}$, the minimization over W for fixed θ does not depend on Ω (Proposition 2), and the profile objective is $\mathbb{E}\|\varepsilon\|_\Omega^2 + \operatorname{tr}(\Omega C(\theta))$ with $C(\theta) = \mathbb{E}[r_\theta r_\theta^\top]$ and $r_\theta = m - \Pi_{\operatorname{span}\phi_\theta}m$. Hence Ω only re-weights the approximation error of the representation across output directions.

B. Synthetic Validation

Table 3 summarizes simulations with known ground truth (AR and VAR windows, Gaussian, t_3 and skewed noise). All numbers come from the released scripts.

A heuristic on the basis (not a theorem). Under skewed noise, an L1 loss targets a median, and an Edgeworth approximation suggests that the gap between the median and the mean of a coordinate grows with its largest loading on a single innovation, which is smaller for Fourier than for time-domain coordinates. In simulations with skewed innovations and H from 4 to 64, the excess MSE of frequency L1 falls from 3.1% to 0.06%, and that of time L1 from 5.1% to about 2.0%.

Table 3. Synthetic checks of Section 3.

Prediction	Result
Theorem 1(a): any $\Omega \succ 0$ gives OLS on a shared design	six Ω (identity, FFT, rFFT, $\hat{\Sigma}_\varepsilon^{-1}$, whitened PCA, random SPD): maximum coefficient difference below 2×10^{-13}
Proposition 3: symmetric noise $\Rightarrow$ same target; skewed $\Rightarrow$ not	$n = 10^5$: Gaussian and t_3 noise $+0.0\%$ for time and frequency L1; centred exponential noise $+3.6\%$ (time L1), $+1.4\%$ (frequency L1)
Corollary 1: SNR weights optimal; GLS only for isotropic signal	SNR weights never worse than MSE, up to -46% risk; GLS -43% when the signal is isotropic, $+11.7\%$ when all SNRs are equal, $+25.9\%$ when signal and noise share the low frequencies
Gains shrink as n/d grows	GLS and rFFT effects shrink monotonically from $\gamma = 0.5$ to 0.025 (GLS -43% to -7%); SNR gains peak at $\gamma = 0.25$ (-46%), then shrink (-28%)
Proposition 4 (i.i.d. design)	formula vs. Monte Carlo within $\pm 2\%$
Sliding windows	AR(1), $\phi = 0.9$: OLS risk ratio 0.97–1.16 to i.i.d. rows with the same marginal covariance, but optimal ridge risk 0.26 vs. 0.71 at $n = 200$ and optimal λ a factor 10 apart
Proposition 5	$n = 1000$: correct rank 3, GLS 0.0199 vs. MSE 0.0204; rank 1–2, GLS worse (0.55 vs. 0.43; 0.17 vs. 0.10); at $n = 40$–80 GLS is worse even with the true Σ_ε
L1 in finite samples (linear, exact fits)	Gaussian noise: L1 worse ($+6$ to $+11\%$ at $n = 100$); t_3: no significant difference; skewed: persistent bias ($+2.9\%$ for time L1 at $n = 10^4$)

The heuristic does not explain the real-data results, where the frequency basis is worse than the time domain on DLinear (Section 5.2).

C. Linear Models Trained Like Deep Models

To separate what a loss estimates from how well it is optimized, we train the linear model class of the closed-form reference with Adam under the FreDF official protocol and each loss. In this class, L1-type losses act as robust M-estimators. Relative to the exact MSE solution, robust losses are better by up to 2.0% on ETTm1, on validation and test data alike. On Weather, time L1 is 5.9% and frequency L1 1.7% worse, which is consistent with a skew-induced bias that the frequency basis reduces. Per channel (official protocol, three seeds), time L1 raises the test MSE of the precipitation channels by 13–14% and of the wind channels by 7–9%, and lowers that of the radiation channels by 11–13%. The official protocol itself leaves the MSE fit 0.3–2.2% above the exact solution (ETTh1 1.6%, ETTh2 2.2%).

Centring (Figure 3, left). We train the same model with and without lookback-mean centring, adding a bias in the uncentred case. This gives a DLinear-like model, since the moving-average decomposition of DLinear is itself linear. The runs cover origins 0–4, 4 horizons and 2 seeds. Relative to the MSE of the same model, L1-type losses gain 0.4–1.7% with centring and 1.3–11.6% without. In absolute terms, the uncentred model with L1 is close to the centred model with MSE (ETTh2: 0.389–0.394 vs. 0.387; ETTm2: 0.259–0.261 vs. 0.264).

D. Selecting the Loss on Validation Data

Validation and test by origin. With the standard validation period, the loss chosen on validation data beats the MSE on test in 80–100% of the cells on iTransformer, close to always using FreDF, but in 40% and 60% of the cells on DLinear for ETTh1 and ETTm1 (Figure 4b). Validation periods in summer and autumn (origins 0–2) favour the MSE, whereas the winter test periods favour L1; once the validation period covers winter, validation and test agree (Figure 4a).

Rolling validation. Pooling all out-of-sample windows observed before the test start across origins lowers the DLinear win rates on ETTh1 and ETTm1 to 30% and 50%, because it adds earlier summer windows. At origins 0–2, no out-of-sample window from the test season exists at all.

Same-season validation (Figure 4c). The DLinear-like model of Appendix C reproduces the validated-selection failure of DLinear origin by origin. On ETTh1, the selected loss beats the MSE at 0, 0, 0, 4 and 4 of the 4 horizons at origins 0–4, and on ETTm1 at 2, 1, 1, 4 and 4. The validation block is the test period shifted back 52 weeks, with the same

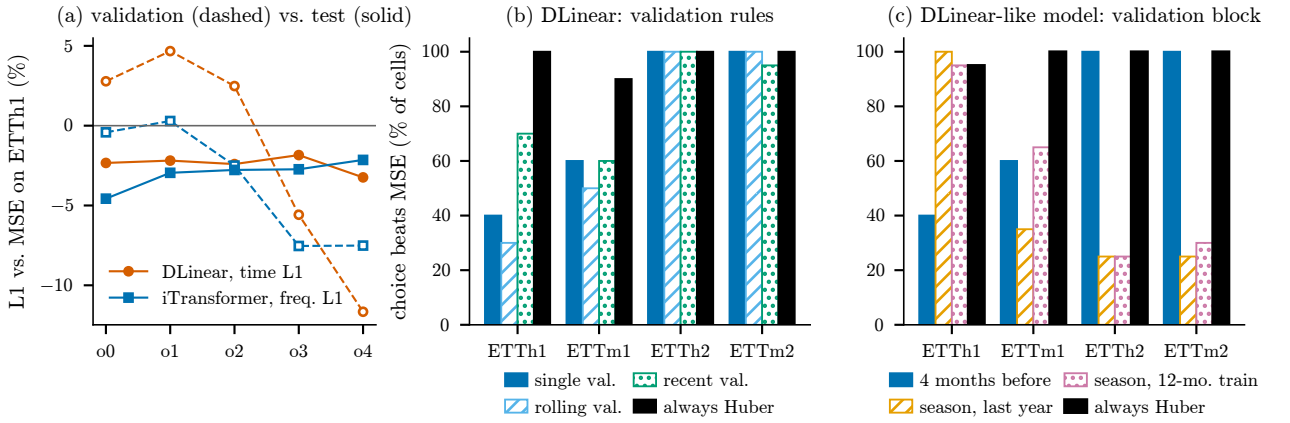


Figure 4. Validated loss selection. (a) Validation (dashed) and test (solid) change of an L1 loss over MSE on ETTh1, by origin; validation periods end in Oct.–Feb. for origins 0–4. (b) DLinear: share of origin × horizon cells in which the loss chosen on validation data beats MSE on test, for three validation rules and for always using time Huber. (c) DLinear-like linear model: the same share when the validation block is the preceding four months, the same season one year earlier, or the same season with training restricted to the standard 12 months.

length. The variant *season* trains on every window whose target ends before the test start and that does not touch the block, while *season_train12* keeps only the standard 12-month training windows. Both use the scaler and the test windows of the standard split. The win rates of the validated choice under the standard, season and season_train12 protocols are 40/100/95% on ETTh1, 60/35/65% on ETTm1, 100/25/25% on ETTh2 and 100/25/30% on ETTm2. We measure the level offset of a period as the mean over channels of the absolute period mean, in standard deviations of the training scaler. It is 0.35–0.36 for the block, which lies inside the training span, 0.38 (ETTh1, ETTm1) and 0.94 (ETTh2, ETTm2) for the standard validation period, and 0.47 and 1.0 for the test period. The block therefore cannot reveal the forward level drift that, judging from Section 5.3, accounts for most of the L1 gain on ETTh2 and ETTm2. The block also removes the same-season data from training, and we did not run a variant that refits on all data after selection. Always using the time Huber loss beats the MSE on average for every dataset and validation design of this model, in 65–100% of the cells.

E. Experimental Details and Reproduction

Code. We use the official repositories at the following commits: FreDF `ff7c571`, Time-o1 `c93d4c5` together with the first-release code `19fbf55` (the later rewrite changes the results although the scripts stay the same), iTransformer `c2426e6`, LTSF-Linear `0c11366`, PatchTST `204c21e` and QDF `eb0693a`. Our patches add loss variants, `--border_offset` for rolling origins, `--use_norm` for iTransformer and `--revin` for DLinear. With their default values, the patched code reproduces the official runs bit for bit.

Protocol. The FreDF scripts (iTransformer, DLinear) use $L = 96$, halve the learning rate every epoch, train for 10 epochs with patience 3, and stop early on the validation MSE. The Time-o1 scripts (Section 5.1 only) also use $L = 96$ and the validation MSE, and train Fredformer for up to 100 epochs (patience 10) on ETT and FreTS for 10 epochs on Weather. Rolling origins shift the 12/4/4-month ETT borders by $k \times 30$ days. For equal-budget tuning, we cross learning rate $\times\{1, \frac{1}{4}\}$ with a halving or a constant schedule (the constant schedule runs for 20 epochs with patience 5). We select the configuration per dataset, origin, horizon and loss by the validation MSE of one seed and re-run it with the other two.

Reproduction of published numbers. We report reproduced and published numbers separately. Under each paper's own protocol, 89–100% of the horizon cells reproduce within $\pm 3\%$ for PatchTST (96%), first-release Time-o1 (100%), QDF (96%), FreDF (89%) and iTransformer (89%), but only 68% for DLinear (LTSF-Linear, $L = 336$), whose three-seed means exceed the published values by 5.0–5.9% on ETTh1, ETTh2 and ETTm2; seed variance explains only part of this gap, and we did not find its cause (the DLinear runs of Section 5 use the FreDF repository with $L = 96$). Along the way, we noticed that the official PatchTST evaluation (Nie et al., 2023) drops the last incomplete test batch and that the MSE baseline of the same backbone differs across papers (0.258, 0.281 and 0.269 for iTransformer on Weather).

Assets and licenses. ETDataset (Zhou et al., 2021) is released under CC BY-ND 4.0, and we use it unmodified. Weather was recorded by the weather station of the Max Planck Institute for Biogeochemistry in Jena,[1] whose documentation states that all its data are freely available for download; we found no explicit license. We use the preprocessed copy of Wu et al. (2021). The code of FreDF, Time-o1, QDF and iTransformer is released under MIT, and that of LTSF-Linear and PatchTST under Apache-2.0. We release patches rather than modified copies, and a download script with checksums rather than the data.

Compute. The experiments of Section 5 and Appendices G–J ran on two internal servers, each with 8 NVIDIA A800 80GB GPUs and two Intel Xeon Platinum 8358 CPUs, with three jobs per GPU. They comprise 5,089 training runs and about 450 job-hours, or roughly 150 GPU-hours. The reproduction of published numbers took 788 runs and about 150 job-hours; 615 of these runs used a third server with NVIDIA H200 GPUs. The per-window errors come from re-evaluating 3,360 checkpoints on CPU, which reproduces the overall test MSE to 10^{-5}. The training-window analysis re-evaluates 3,401 checkpoints on CPU, and the interventions of Table 7 train 3,520 linear models on CPU. The same job gives the same test MSE to 10^{-5} on A800, A100 and H200 GPUs. The released script `scripts/release/compute_budget.py` computes these totals from the run records.

F. Diebold–Mariano Tests

For each cell, we average the per-window test MSE over common seeds and test $d_t = e_{1t} - e_{2t}$. The long-run variance is the equal-weighted cosine estimator $\hat{\omega}^2 = \frac{1}{B}\sum_{j=1}^{B} \Lambda_j^2$, $\Lambda_j = \sqrt{2/T}\sum_t d_t \cos(\pi j(t-\frac{1}{2})/T)$, with $B = \max(1, \lfloor T/2H \rfloor)$ and t_B critical values. Overlapping H-step windows make d_t strongly dependent up to lag H. In simulations of H-window averages of white or AR(1) ($\phi = 0.95$) noise with $H \in \{96, 336, 720\}$ and the test lengths of ETTh, this test rejects a true null in 5–7.5% of replications at the nominal 5%, whereas a Newey–West estimator with bandwidth H and normal critical values rejects in 14–35%. We apply Holm's correction within each backbone $\times$ dataset $\times$ comparison family of 20 origin $\times$ horizon cells. Power is low at long horizons ($B = 1$ at $H = 720$). Because the test compares seed-averaged forecasts, it conditions on the trained models and does not include training randomness; Section 5.2 reports how often single training runs agree.

G. Additional Results

Table 4. The comparisons of Table 2 in MSE and in MAE: paired relative change of the test error (%), mean over origins of the horizon $\times$ seed means, shown as MSE / MAE.

Backbone	Comparison	ETTh1	ETTh2	ETTm1	ETTm2
iTransformer	FreDF vs. MSE	−3.0 / −2.6	−3.7 / −2.7	−2.0 / −2.2	−5.4 / −4.5
iTransformer	frequency L1 vs. MSE	−3.0 / −2.7	−3.8 / −2.9	−2.0 / −2.2	−5.5 / −4.6
iTransformer	time L1 vs. MSE	−1.5 / −2.3	−2.4 / −2.4	−1.9 / −4.1	−4.2 / −4.1
iTransformer	time Huber vs. MSE	−1.3 / −2.1	−2.4 / −2.3	−1.7 / −3.6	−4.1 / −3.7
iTransformer	frequency vs. time L1	−1.5 / −0.3	−1.4 / −0.5	−0.1 / +2.0	−1.4 / −0.6
DLinear	frequency L1 vs. MSE	+1.9 / +0.3	−12.9 / −8.8	+0.5 / −0.3	−9.0 / −8.9
DLinear	time L1 vs. MSE	−2.4 / −2.5	−14.3 / −10.3	−1.0 / −3.8	−9.8 / −10.4
DLinear	time Huber vs. MSE	−2.4 / −2.4	−13.8 / −9.7	−1.3 / −3.5	−10.3 / −9.6
DLinear	frequency vs. time L1	+4.4 / +2.9	+1.5 / +1.7	+1.5 / +3.6	+0.8 / +1.6

MAE. Table 4 repeats the comparisons of Table 2 in MAE. The L1-type losses gain in MAE as they do in MSE. The only change of ranking is frequency versus time L1 on iTransformer/ETTm1 (-0.1% in MSE, $+2.0\%$ in MAE). On the standard split, the L2 variants do not improve the MAE either (FFT-L2 0.0% and $+0.1\%$, rFFT-L2 $+0.6\%$ to $+3.3\%$, PCA-L2 0.0% to $+2.2\%$).

Tuned MSE on the standard split. We cross learning rate $\times\{1, \frac{1}{4}\}$, a halving or a constant schedule, and dropout $\{0.1, 0.3\}$ for iTransformer, select one configuration per dataset, horizon and loss by the validation MSE of one seed, and re-run it with the other two. The official configuration is never selected. Relative to the tuned MSE, the official MSE

[1] https://www.bgc-jena.mpg.de/wetter/

changes the test MSE by -0.2, $+1.2$, $+1.6$ and $+2.4\%$ on ETTh1, ETTh2, ETTm1 and Weather, official FreDF by -4.8, -3.0, -2.6 and -0.6%, frequency L1 by -4.9, -3.1, -2.5 and -2.3%, time L1 by -2.2, -0.9, -3.0 and -3.3%, and tuned FreDF by -3.7, -3.3, -3.5 and -0.3%. We did not tune FreDF's mixing weight λ. Its official value (0.2, 0.1, 0.2 and 0.4) and $\lambda = 0$, i.e. frequency L1, differ by at most 0.15 percentage points on the ETT datasets, on the standard split and on average over the rolling origins, whereas on Weather frequency L1 gains 4.5% against 2.7% for FreDF.

Sub-periods (standard split). On four consecutive validation and four test parts (about one month each on ETT, 9–18 days on Weather), time L1 improves on the MSE in 7, 5, 8 and 7 of the 8 parts on ETTh1, ETTh2, ETTm1 and Weather, and frequency L1 in 7, 8, 6 and 8, compared with 2, 0, 0 and 0 for frequency L2 and 4, 2, 0 and 0 for PCA-L2. All L1 variants gain more on the test than on the validation parts. Against time L1, frequency L1 changes the test MSE in the four ETTh1 test months by -2.9, -3.8, -5.4 and $+4.4\%$. Because the sign flips in the last month, conclusions about the basis from a single test period depend on its months.

Distance to the exact linear MSE solution. Relative to the closed-form reference, the test MSE of iTransformer trained with the MSE ranges across origins from $+0.4$ to $+4.4\%$ on ETTh1, $+2.3$ to $+8.0\%$ on ETTh2, 0.0 to $+1.4\%$ on ETTm1 and $+3.3$ to $+6.4\%$ on ETTm2. With FreDF, the ranges are -1.8 to -0.4%, -0.6 to $+3.6\%$, -4.2 to $+0.4\%$ and -2.1 to $+2.0\%$. The reference is the optimum of its own class and not a bound for other classes; for example, DLinear beats it on ETTm1 at every origin.

Horizon. Averaged over the four ETT datasets, the gain of FreDF on iTransformer shrinks with the horizon (-4.3, -3.9, -3.3 and -2.6% for $H = 96, \ldots, 720$), while the gain of time L1 on DLinear grows (-4.3, -5.9, -7.8 and -9.5%). This is consistent with level errors of an unnormalized model that grow with the horizon.

H. Mechanism Details

Early stopping. The slack of validation-based stopping, that is, the test MSE at the selected epoch minus the best test MSE over the epochs run, is 0.6–0.8% for iTransformer trained with MSE on ETT, 0.1–0.9% with FreDF or frequency L1, and 0.0–0.3% for DLinear. Early stopping can therefore explain at most about 0.8%. Against the MSE at its test-optimal epoch, FreDF changes the test MSE by -2.4, -3.0, -1.2 and -4.8% and time L1 by -1.0, -1.7, -1.1 and -3.6% on ETTh1, ETTh2, ETTm1 and ETTm2.

Where MSE fails. The correlation between the lookback level offset of a window and the excess error of MSE-trained DLinear over its L1 counterpart is -0.08, -0.44, -0.10 and -0.32 on ETTh1, ETTh2, ETTm1 and ETTm2. On the 10% of windows with the largest offset, MSE is worse by 0.5% on ETTh1 and better by 11.6%, 3.8% and 22.4% on the other three datasets; on the other 90%, it is worse by 2.7%, 22.5%, 2.1% and 19.1%.

Gain and level offset. We computed the level offset of Appendix D for each origin after the gains were known. Between origins 0 and 4, the test periods of ETTh2 and ETTm2 move from 1.29 to 0.60 training standard deviations, and those of ETTh1 and ETTm1 stay at 0.43–0.52. The change of test MSE of DLinear goes from -22.3% to -8.4% on ETTh2 and from -16.0% to -3.7% on ETTm2 for time L1, and from -22.0% to -6.3% and from -14.4% to -3.3% for frequency L1. Over the 20 dataset $\times$ origin cells, its Spearman correlation with the offset is -0.88 for time L1 and -0.93 for frequency L1, and -0.82 and -0.86 for iTransformer with its normalization switched off (15 cells). With normalization, the correlation within each dataset is positive for iTransformer (0.3 to 0.7), and DLinear with RevIN stays between -1.9% and $+3.9\%$ at every origin. Overlapping origins whose test periods differ in season limit what these correlations show.

Training windows. We re-evaluate every checkpoint of Table 2 and of the normalization controls on its training windows (CPU; the recomputed test MSE matches the logged one to 10^{-4}). Table 5 compares the training and test windows. Without normalization, the L1 fit is worse than the MSE fit on every subset of training windows, most on those with the largest lookback level offset, and better on the test windows. The share of the training error carried by the largest-error windows hardly depends on normalization.

Direction of the error. Table 6 splits the change of test MSE into the part along the DC direction of the horizon, i.e. the mean level of the forecast error, and the rest. The L1 losses reduce mostly the level error, more than its share of the MSE model's error, and the extra error of the rFFT half-spectrum L2 lies mostly along the DC direction, which it weights twice.

Table 5. Training and test windows (ETTh2, ETTm2; mean over origins × horizons × seeds). Share of the training squared error of the MSE fit carried by its 10% largest-error windows, and the change of the MSE of the time-L1 fit relative to the MSE fit (%) on the 10% of windows with the largest lookback level offset and on the other 90%, on the training and on the test windows.

Model	Dataset	top-10% share of training error	training windows largest offset	training windows other 90%	test windows largest offset	test windows other 90%
DLinear	ETTh2	51%	+16.0	+1.4	+20.2	−17.9
DLinear	ETTm2	56%	+20.8	+0.2	+29.1	−16.0
DLinear + RevIN	ETTh2	52%	+2.5	+0.0	+0.2	−0.6
DLinear + RevIN	ETTm2	59%	+5.3	+0.2	+3.9	−1.2
iTransformer, norm. off	ETTh2	47%	+20.4	+8.6	−25.1	−16.1
iTransformer, norm. off	ETTm2	47%	+27.2	+7.0	−24.7	−15.7
iTransformer	ETTh2	48%	+4.4	+0.6	−6.7	−1.5
iTransformer	ETTm2	55%	+3.1	−5.7	−5.6	−3.7

Table 6. Change of the test MSE relative to the MSE (iTransformer, standard split; % of the MSE model's test MSE, mean over horizons × seeds), shown as total / part along the DC direction of the horizon (the mean level of the forecast error). The Nyquist part is below 0.01 percentage points everywhere. Last row: DC share of the MSE model's own test error.

Loss	ETTh1	ETTh2	ETTm1	Weather
rFFT L2	+0.41 / +0.21	+1.77 / +1.64	+0.69 / +0.34	+3.88 / +2.05
frequency L1	−4.61 / −2.86	−4.17 / −2.38	−4.01 / −2.42	−4.49 / −2.77
FreDF	−4.51 / −2.75	−4.14 / −2.42	−4.11 / −2.52	−2.74 / −1.91
time L1	−1.95 / −0.54	−2.04 / −0.79	−4.50 / −2.34	−5.41 / −3.56
DC share of the MSE error	18%	41%	27%	46%

Interventions in the linear model class. Table 7 trains the linear models of Appendix C with modified squared losses. Without centring, dropping the largest-error or the largest-offset windows of each batch recovers a large part of the L1 gain, per-window gradient clipping at the same quantile recovers less, and global gradient clipping recovers little. Balancing the squared error per horizon step and channel recovers about half of it, also with a running scale that does not down-weight single batches; balancing per frequency recovers less. With centring, all changes are small (at most 1.7%), and the frequency-balanced L2 with a running scale matches frequency L1 on ETTh1 and ETTh2. Doubling the DC weight raises the test MSE in both models, and doubling the Nyquist weight has no effect.

I. Output-Direction Weighting

The SNR plug-in estimates the noise and signal levels of each direction with formulas for i.i.d. rows. On overlapping windows these estimates are biased, which is the case that Proposition 4 does not cover, and the plug-in is worse than uniform ridge by 1.5% on average and by 15% at worst.

J. Efficiency

DLinear has 18.6k–140k parameters, and iTransformer 224k–305k on ETT. Single-thread CPU inference (batch size 1) takes 0.27–0.80 ms per window for DLinear and 1.1–1.2 ms for iTransformer; peak memory is 590–640 MB, mostly the PyTorch runtime. All losses cost the same per epoch (22–25 s for iTransformer on ETTh1, three jobs per A800). Relative to the MSE, an iTransformer job on ETT takes 1.2–1.7 times as long with FreDF and frequency L1, which run more epochs (8.4 and 8.9 against 4.7 on ETTh1), and 1.0–1.1 times with time L1 and Huber (1.4–1.5 on ETTm1); on DLinear the ratio is 0.9–1.6, and on Weather about 0.6, as the L1-type losses stop earlier (5–7 against 10 epochs). Choosing among k losses multiplies training cost by k; inference is unchanged.

Table 7. Training-side interventions in the linear model class (Adam, official protocol; ETT × origins 0–4 × 4 horizons × 2 seeds): paired change of test MSE relative to the MSE of the same model (%). Windows are weighted or dropped at the batch 90% quantile of their error or of their lookback level offset; global clipping caps the gradient norm at the median of the previous norms; balanced losses scale each coordinate by its RMS over the batch or by a running mean over batches.

	DLinear-like (no centring)				centred			
Loss	ETTh1	ETTh2	ETTm1	ETTm2	ETTh1	ETTh2	ETTm1	ETTm2
time L1	−2.6	−11.0	−1.3	−6.3	−0.5	−1.2	−0.6	−0.4
trimmed MSE	−1.1	−7.2	−0.9	−6.8	−0.3	−1.0	−0.5	−1.1
MSE w/o largest-offset windows	−0.9	−6.2	−0.4	−6.5	−0.2	−0.5	−0.1	−1.0
per-window gradient clipping	−0.2	−2.9	−0.3	−4.2	−0.1	−0.8	−0.1	−0.9
global gradient clipping	−0.1	−1.0	−0.1	−1.8	+0.0	−0.4	−0.0	−0.7
time L2, batch-balanced	−1.0	−6.7	−0.7	−5.7	−0.2	−0.5	−0.4	−0.5
time L2, balanced (running)	−0.7	−4.9	−0.4	−3.8	−0.1	−0.2	−0.2	−0.2
rFFT L2, batch-balanced	−2.2	−3.9	−0.5	−4.9	−0.9	−1.7	−0.1	−0.8
rFFT L2, balanced (running)	−2.2	−1.8	−0.4	−2.1	−1.0	−1.7	−0.1	−0.5
DC weighted twice	+1.0	+0.6	+0.5	+0.6	+0.5	+1.0	+0.3	+0.3
Nyquist weighted twice	−0.0	−0.0	−0.0	−0.0	−0.0	−0.0	−0.0	−0.0

Table 8. Closed-form linear reference, 168 settings; change of test MSE relative to uniform ridge.

Scheme	Mean	Better in	Worst / best
Per-direction λ, time steps	−0.00%	60%	+0.9% / −0.4%
Per-direction λ, Fourier	−0.01%	59%	+3.0% / −1.7%
Per-direction λ, residual PCA	+0.07%	57%	+7.3% / −1.7%
GLS ($\hat{\Sigma_\varepsilon}^{-1}$)	+0.13%	40%	+4.4% / −1.2%
Label-PCA truncation	+0.00%	–	(almost never truncates)
SNR plug-in (Corollary 1)	+1.54%	18%	+15.0% / −0.1%